Eran Rehani

Physics Student · Hebrew University of Jerusalem

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Third-year Extended Single-Major B.Sc. Physics student at the Hebrew University of Jerusalem, working in computational physics and scientific programming. I write simulations from first principles (Monte Carlo, molecular dynamics, numerical quantum mechanics), analyse the output in Python, and report the results in LaTeX.

EDUCATION

B.Sc. Physics, Extended Single Major · Hebrew University of Jerusalem — 2023–

Completed Year 3. Coursework listed below; LaTeX summaries of the Year 3 courses are published at lecture-notes-huji.

EXPERIENCE

Personal Trainer · Certified, Wingate Institute — 2021–present

Certified Personal Trainer, Wingate Institute. Built training programmes, tracked client progress, and managed my own client book, first at a gym and then independently.
Still training one shift a week alongside full-time studies.

MILITARY SERVICE

Intelligence Corps, IDF · Cyber & Information Security — 12/2018–5/2021

Worked in Linux and networking environments; wrote Python and Bash tooling to automate recurring analysis. Owned a technical domain end to end and trained incoming personnel on it.
Handled live security incidents on an on-call rotation, under time pressure and with sensitive data.

IAF Pilot Course, Selected Candidate · Ben-Gurion University — 7/2018–12/2018

Passed national Israeli Air Force screening and entered the pilot course; completed the academic coursework Ben-Gurion University delivers to the cohort.

TECHNICAL SKILLS

Programming	Python, MATLAB, LaTeX
Libraries	NumPy, SciPy, Matplotlib, Numba (JIT), OpenCV, multiprocessing
Tools	Git, Linux, Jupyter, VS Code, AI-assisted development workflows
Languages	Hebrew (native), English (high level, university exemption); writes technical reports in English

RELEVANT COURSEWORK

Physics: Mechanics & Special Relativity, Electromagnetism, Analytical Electrodynamics, Analytical Mechanics, Waves & Optics, Thermal Physics, Statistical Physics, Quantum Mechanics 1–2, Approximation Methods, Nuclear Physics, Particle Physics, Continuum Mechanics, Labs A–B. **Mathematics:** Linear Algebra, Calculus, Probability, Complex Functions, Equations of Mathematical Physics. **Computational:** Computational Physics (Python), MATLAB.

PROJECTS & LAB WORK

Chaos & Nonlinear Dynamics · Python · NumPy · Matplotlib — 2024

Mapped chaos in a dripping faucet (bifurcation diagrams, return maps, surface tension) and in an RLD circuit (I–V curve, diode capacitance, onset of chaos), then measured the Feigenbaum constant from the bifurcation diagram. [drip report] [RLD report]

Monte Carlo & Molecular Dynamics Simulations · Python · Numba · multiprocessing — 2024

2D Ising model with heat-bath (Glauber) dynamics and Numba JIT. Recovered the Onsager critical temperature T_c and the exponents β, γ, ν by finite-size scaling.
Hard-Disk Gas MD: exact event-driven molecular dynamics. Verified the Maxwell–Boltzmann distribution and the radial distribution function $g(r)$.

Numerical Schrödinger Solver · Python · NumPy · SciPy — 2024

Solved the radial Schrödinger equation with finite differences and the Numerov method ($O(h^4)$), checked convergence against the analytic hydrogen spectrum, added Coulomb corrections, and built a self-consistent-field solver for multi-electron atoms.

Computational Physics (77315) · Python · Jupyter — 2026

Floating-point stability, interpolation, numerical integration and root-finding, applied to the structure of white dwarf stars.

Additional: Einstein Solid MC · Brownian Motion Tracking · Photoelasticity · RLC Frequency Response · Laplace Solver